

DAY 4 – NETWORK CABLE TYPES AND TOOLS

Course: Diploma in Cyber Security

**Topic: Network Cables (Ethernet vs
Fiber) & Crimping Tools**

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TASK 1 – ETHERNET VS FIBER OPTIC COMPARISON

- **1. Material & Signal:**

Ethernet cables use copper wires to send electrical signals, while fiber optic cables use thin glass fibers to send light signals.

- **2. Distance:**

Ethernet works well only up to 100 meters, but fiber cables can carry data over very long distances like 10 km to 40 km without losing signal.

- **3. Interference:**

Ethernet can get disturbed by electrical interference from nearby power lines, whereas fiber optics is not affected by any electrical noise.

Real-World Examples:

- Ethernet Cable: Used at home to connect my Wi-Fi router to my laptop or smart TV for daily internet use.
- Fiber Optic Cable: Used across a college campus to connect different department buildings together with a central server room.

TASK 2 – T568B STRAIGHT-THROUGH WIRING DIAGRAM & COLOR ORDER

To make a standard straight-through network cable, we use the T568B wiring standard. The 8 internal wires go into the RJ-45 connector in this exact order:

- **Pin 1:** White / Orange
- **Pin 2:** Orange
- **Pin 3:** White / Green
- **Pin 4:** Blue
- **Pin 5:** White / Blue
- **Pin 6:** Green

- **Pin 7:** White / Brown
- **Pin 8:** Brown

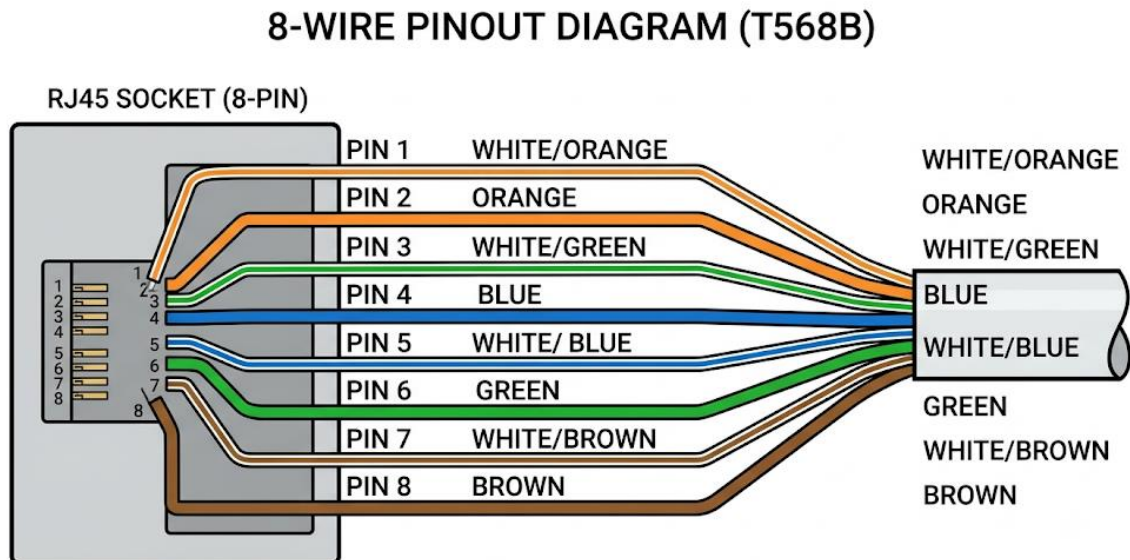


Figure 1: Wire color sequence for T568B standard inside an RJ-45 connector.

TASK 3 – HOW TO CRIMP AN RJ-45 CONNECTOR ON CAT6 CABLE

1. Cut the Cat6 cable to the size I need.
2. Carefully remove about 2 cm of the outer plastic jacket using the wire stripper.
3. Untwist all the wire pairs and cut off the center plastic piece.

4. Line up all 8 wires flat in the T568B order: *White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.*
5. Cut the tips of the wires straight so they are all the same length.
6. Slide the wires all the way into the RJ-45 connector, checking that every wire hits the end.
7. Place the connector into the crimper tool and press down hard to lock the pins.
8. Plug both ends of the cable into a LAN tester to check that all lights from 1 to 8 blink properly in order.

Two Mistakes to Avoid:

- Mixing up the wire colors, because the cable will not work if the order is wrong.
- Not pushing the wires deep enough into the connector before pressing the crimper.

TASK 4 – GAMING CAFÉ NETWORK SETUP (10 PCs)

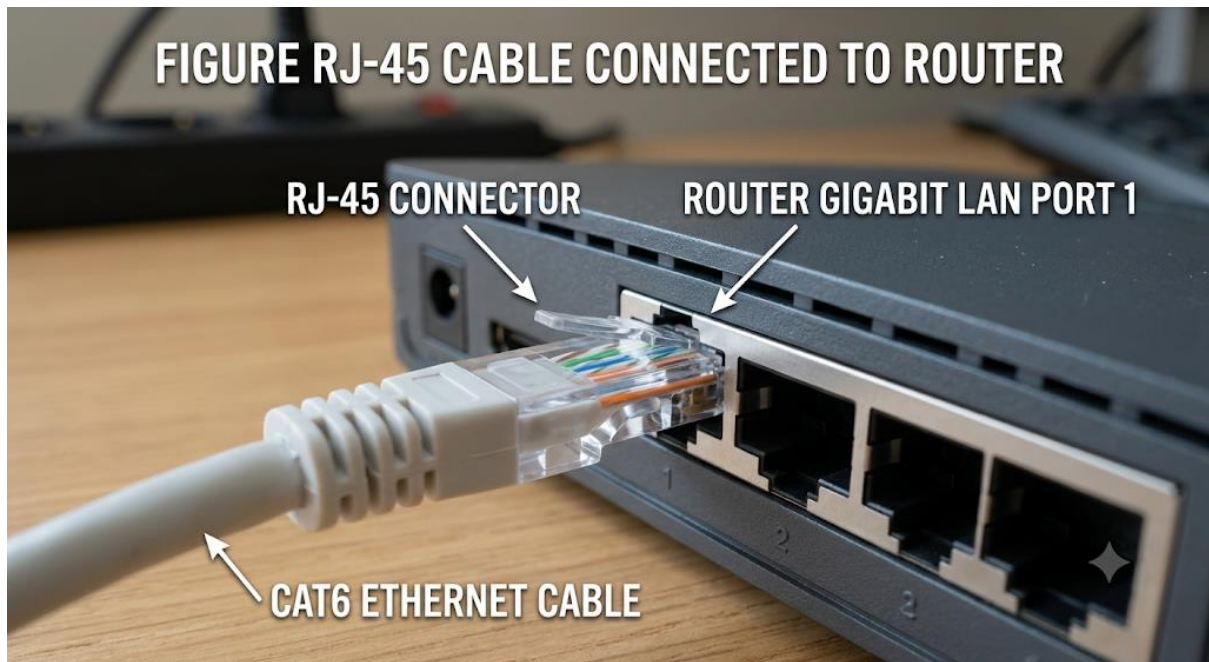
- **My Cable Choice:** Cat6 Ethernet Cable
- **My Tool Choice:** RJ-45 Crimping Tool

Why I Chose This:

"For a 10-PC gaming café, all computers are inside one room close to each other. Cat6 Ethernet cable is cheap, easy to make using an RJ-45 crimper, and gives 1 Gbps speed with almost zero lag for gaming. Fiber optics is not needed here because the distance is short, and fiber tools are too expensive and hard to fix if a cable breaks."

TASK 5 – NETWORK CABLE PHOTO EVIDENCE

- **Cable Found:** Cat6 Ethernet Cable (Yellow/Blue LAN cord).
- **Connector:** Plastic RJ-45 connector.
- **Where It Is Connected:** From the back of the home Wi-Fi router directly into the PC.
- **Why It Is Used:** To get direct and steady internet speed without Wi-Fi drop issues.



CONCLUSION

In this practical, I learned about different network transmission media and how to make a standard Ethernet cable. I understood the practical differences between copper Ethernet cables and fiber optics in terms of speed, distance, and cost. I learned how to arrange the 8 colored wires using the standard T568B order, cut them evenly, and crimp them into an RJ-45 connector using a crimping tool. I also realized the importance of checking connections with a LAN tester and avoiding mistakes like wrong color order or loose wire insertion. Finally, through the gaming setup scenario, I understood why Cat6 copper cabling is the most reliable, budget-friendly, and practical choice for short-distance local networks.